

Beyond the Swipe: Predicting Dating App Match Outcomes

Group Project Report

Course: WIA1006 Machine Learning

Occurrence: 2

Group: Group 5 — Team TrueLove

Project Resources:

- **Presentation Video:** [Google Drive Video](#)
- **Google Colab Notebook:** [Colab Notebook](#)
- **Streamlit Dashboard Demo:** [Streamlit Dashboard](#)

Team Member	Matric Number	Technical Responsibility
Tan Jun Yong	25005715	Project coordination, pipeline architecture, model integration
Ng Hong Jon	25005754	Data preprocessing, feature encoding, exploratory data analysis
Loo Jiun Wei	25006149	Random Forest modelling, GridSearchCV tuning, results interpretation
Tan Kang Zheng	25005697	SVD/t-SNE visualization, metric evaluation, error analysis
Tan Sze Yung	25005884	Streamlit dashboard prototype, deployment testing, ethical discussion

1. Problem and Objective

This project investigates whether dating-app user data can predict match_outcome. The task was designed as a supervised 10-class classification problem, using the original target labels rather than simplifying them into fewer categories.

The ten target classes were: Blocked, Catfished, Chat Ignored, Date Happened, Ghosted, Instant Match, Mutual Match, No Action, One-sided Like, and Relationship Formed.

The dataset contains 50,000 synthetic user records, 24 predictor features, and one target variable. Features include demographic information, app behaviour, swipe patterns, interaction signals, interest tags, and relationship preferences.

The main objective was not only to build predictive models, but also to test whether the synthetic dataset contains meaningful learnable signal. Since the target classes are nearly balanced, the expected chance-level performance is approximately 10% accuracy.

2. Methodology and Model Explanation

The dataset was inspected before modelling. It contained no missing values and no duplicate rows, so no imputation was required.

The preprocessing pipeline was based on feature type:

Feature Type	Examples	Transformation
Numerical	age, app usage time, likes received, mutual matches	StandardScaler
Categorical	gender, location type, income bracket, relationship intent	OneHotEncoder
Multi-value text	interest_tags	CountVectorizer

The data was split using an 80/20 train-test split with a fixed random state for reproducibility. The target column was removed before preprocessing to avoid target leakage.

Six machine learning models were evaluated:

1. **Logistic Regression**
2. **Decision Tree**
3. **Random Forest**
4. **K-Nearest Neighbours (KNN)**
5. **Support Vector Machine (SVM)**
6. **Gradient Boosting**

After the initial comparison, GridSearchCV was used to tune Random Forest hyperparameters. Random Forest was selected for tuning because it is a strong non-linear ensemble model suitable for mixed structured data.

3. Results and Visualization

3.1 Model Performance Summary

Model	Accuracy	Weighted F1	Interpretation
Dummy Baseline	10.22%	0.0190	Majority-class benchmark
Logistic Regression	9.94%	0.0991	Near chance level
Decision Tree	9.43%	0.0943	Near chance level
Random Forest	9.90%	0.0986	Near chance level
KNN	9.55%	0.0903	Near chance level
SVM	9.61%	0.0959	Near chance level
Gradient Boosting	9.61%	0.0950	Near chance level
Tuned Random Forest	10.15%	0.1009	Best tuned model, still near baseline
FLAML AutoML	9.73%	0.0741	AutoML fallback; below tuned Random Forest

The Dummy Baseline achieved 10.22% accuracy, which is expected because the dataset has ten nearly balanced classes. The best tuned model, Random Forest with GridSearchCV, achieved only 10.15% accuracy. This is not a meaningful improvement over chance-level prediction.

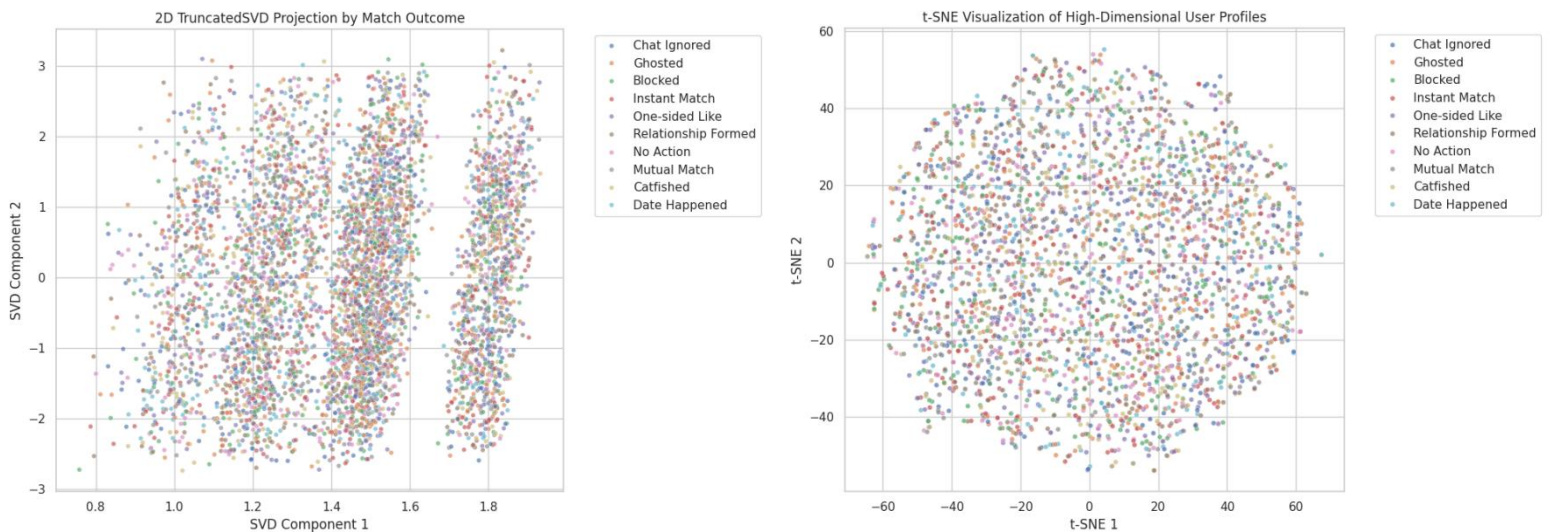
The confusion matrix showed no strong diagonal pattern, meaning the model did not consistently predict the correct class. Class-level precision, recall, and F1-scores were also close to 0.10, confirming that no outcome class was predicted reliably.

3.2 SVD and t-SNE Visualization

Dimensionality reduction was used to inspect whether the transformed feature space contained visible class separation.

TruncatedSVD showed heavy overlap among the ten classes. t-SNE also showed mixed clusters with no clear boundary between match outcomes.

These visualizations support the metric results: the 24 features do not form separable patterns for the ten target classes.



4. Comparison to Auto-Sklearn and AutoML Fallback

Auto-sklearn was treated as the requested AutoML benchmark. In the available runtime, the autosklearn package could not be imported, so the notebook recorded this limitation transparently and used FLAML AutoML as a practical fallback under the same train-test split and preprocessing design.

The FLAML fallback achieved 9.73% accuracy and a weighted F1-score of 0.0741, which was below the tuned Random Forest result. This supports the main conclusion that the dataset has weak feature-target signal rather than a problem that can be solved simply by switching to a more automated model search.

AutoML can optimise models and hyperparameters, but it cannot create predictive information that is absent from the data.

	Model	Accuracy	Precision	Recall	F1 Score
0	Tuned Random Forest	0.1015	0.101786	0.1015	0.100858
1	Logistic Regression	0.0994	0.099492	0.0994	0.099070
2	Random Forest	0.0990	0.099585	0.0990	0.098636
3	SVM	0.0961	0.096386	0.0961	0.095949
4	Gradient Boosting	0.0961	0.095511	0.0961	0.095025
5	Logistic Regression + TruncatedSVD	0.0953	0.095299	0.0953	0.094556
6	Decision Tree	0.0943	0.094377	0.0943	0.094295
7	KNN	0.0955	0.095332	0.0955	0.090305
8	FLAML AutoML	0.0973	0.088925	0.0973	0.074128
9	Dummy Baseline	0.1022	0.010445	0.1022	0.018953

5. Insights and Interpretation

The key finding is that the synthetic dataset does not contain enough learnable structure to predict `match_outcome`.

Although the features appear realistic, such as swipe-right ratio, likes received, message count, mutual matches, and relationship intent, they do not meaningfully explain the target labels. This is shown by:

Evidence	Finding
Target setup	10 nearly balanced classes
Chance benchmark	Approximately 10%
Dummy Baseline	10.22% accuracy
Best tuned model	10.15% accuracy
Model comparison	All six models near chance
SVD/t-SNE	No clear class separation
Confusion matrix	No strong diagonal pattern

The project therefore demonstrates the “Garbage In, Garbage Out” principle. Model complexity cannot overcome weak or random feature-target relationships. If the synthetic target labels were not generated from meaningful behavioural patterns, machine learning models cannot reliably predict them.

This is a valid scientific result. The low accuracy does not indicate poor implementation; it shows that the dataset itself is unsuitable for reliable prediction.

Streamlit Dashboard Prototype

A Streamlit dashboard prototype was developed to demonstrate deployment. Users can enter profile, behaviour, interaction, and preference details. The dashboard then displays model-assigned probabilities for the ten match outcomes.

However, because the model performs near chance level, the dashboard is clearly labelled as an educational prototype, not a real dating advice system. This is important because relationship outcomes involve complex human factors that are not captured by surface-level app data.

Output正确性验证

OBS 录制窗口教程

ML_Assignment_Submission_Re

Match Outcome Predictor - Stre

+

Ask Gemini

my-app-dashboard-9outn4irtvholrczsdhcv.streamlit.app

Install

Fork

Match Outcome Predictor

Fill in a dating profile below and see what the model predicts will happen.

All fields mirror the training data — the more accurate your inputs, the more meaningful the prediction.

Demographics

Age

1828

60145210

Gender

Male

Height (cm)

170

145130.00

Weight (kg)

70.00

Body type

Slim

Zodiac sign

Aries

Background

Sexual orientation

Income bracket

Relationship intent

Output正确性验证

OBS 录制窗口教程

ML_Assignment_Submission_Re

Match Outcome Predictor - Stre

+

Ask Gemini

my-app-dashboard-9outn4irtvholrczsdhcv.streamlit.app

Install

Fork

Background

Sexual orientation

Straight

Income bracket

Very Low

Relationship intent

Serious Relationship

Location type

Urban

Education level

No Formal Education

Interest tags (pick 1-5)

Fitness

Traveling

App Behaviour

Daily app usage (min)

090

3000.001.00

Swipe-right ratio

0.45

Last active hour (0-23)

21

023

Usage category

Moderate User

Swipe style

Moderate

Profile photos

3

16

Peak swipe time

Evening

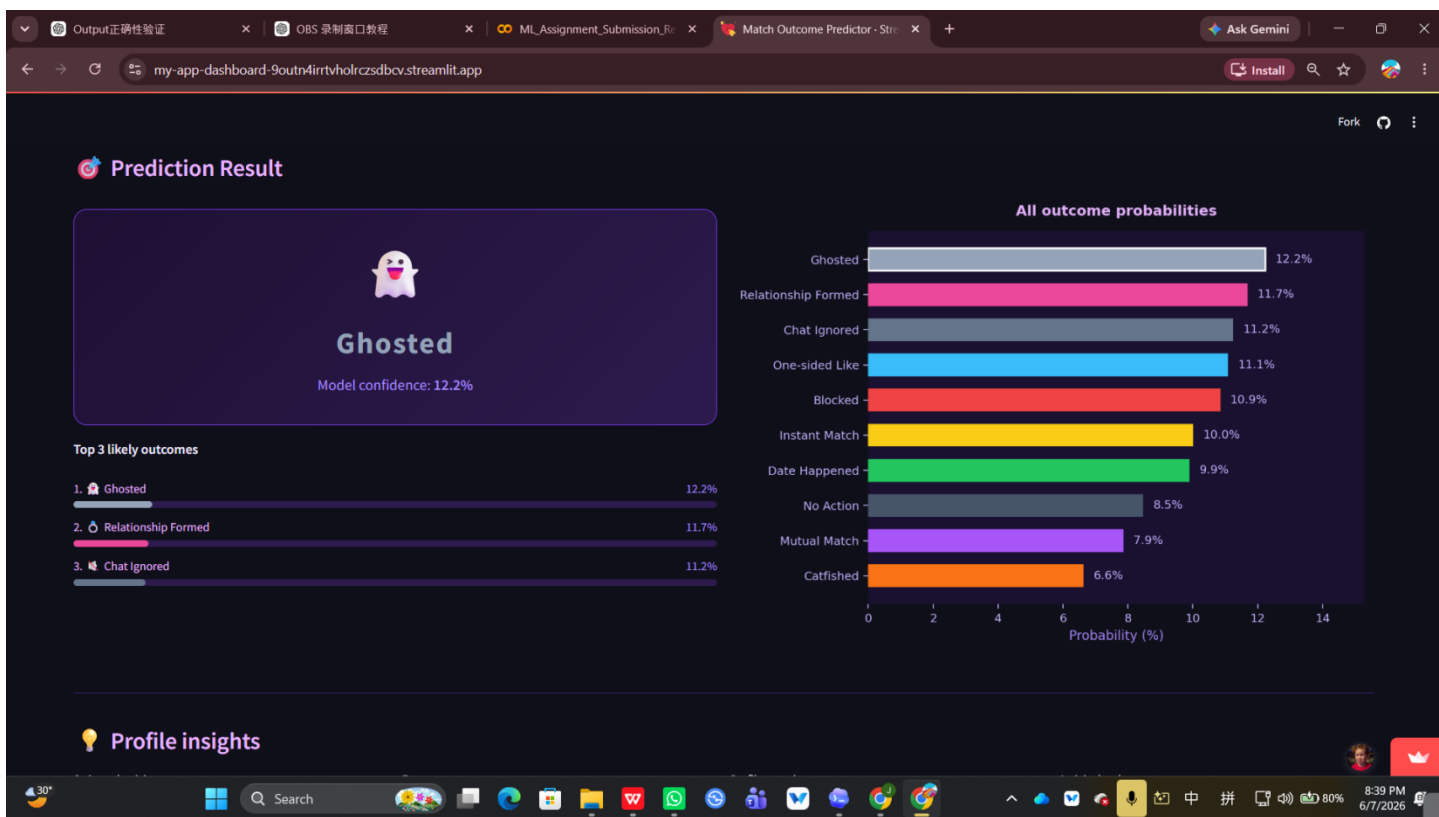
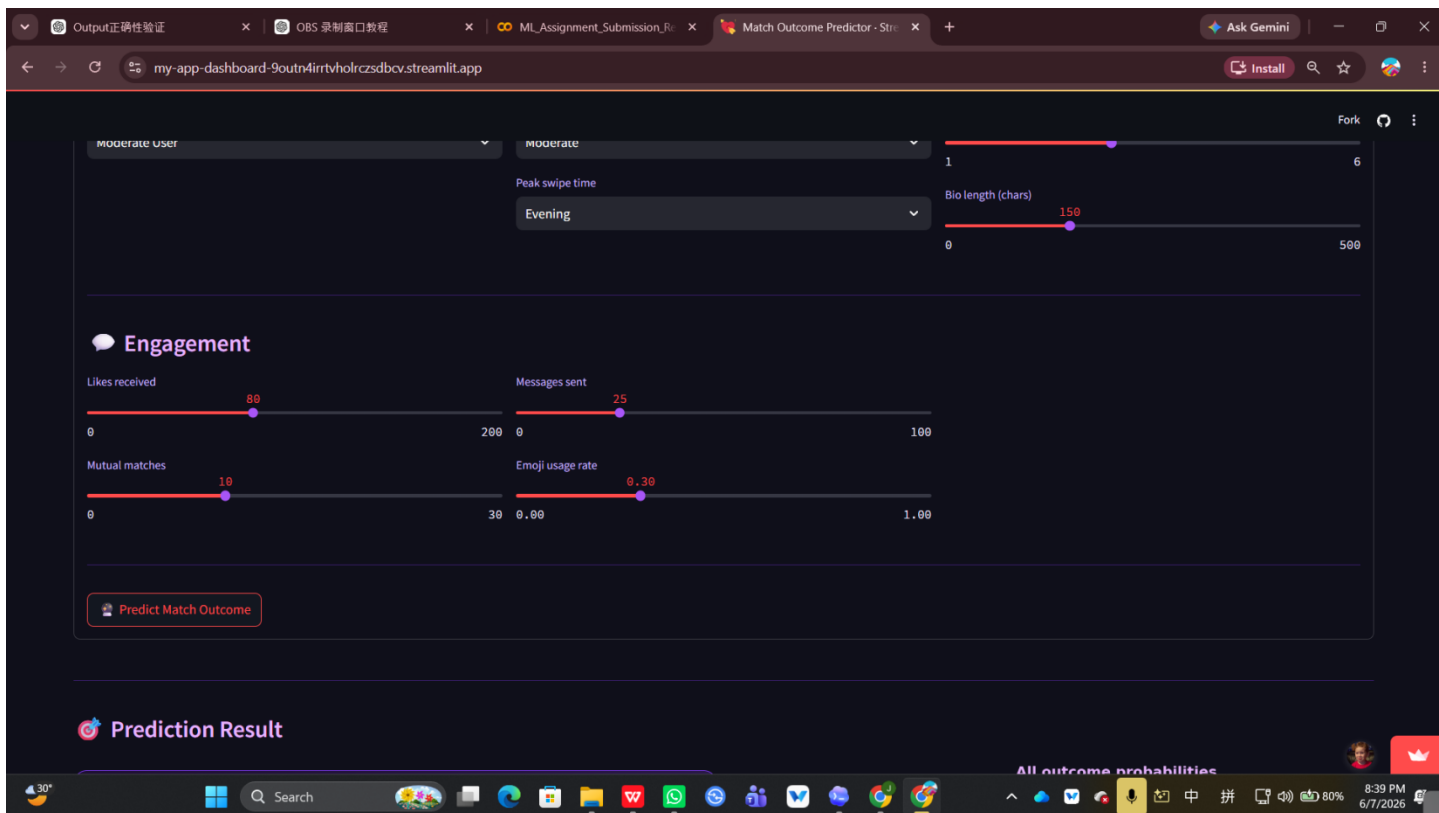
Bio length (chars)

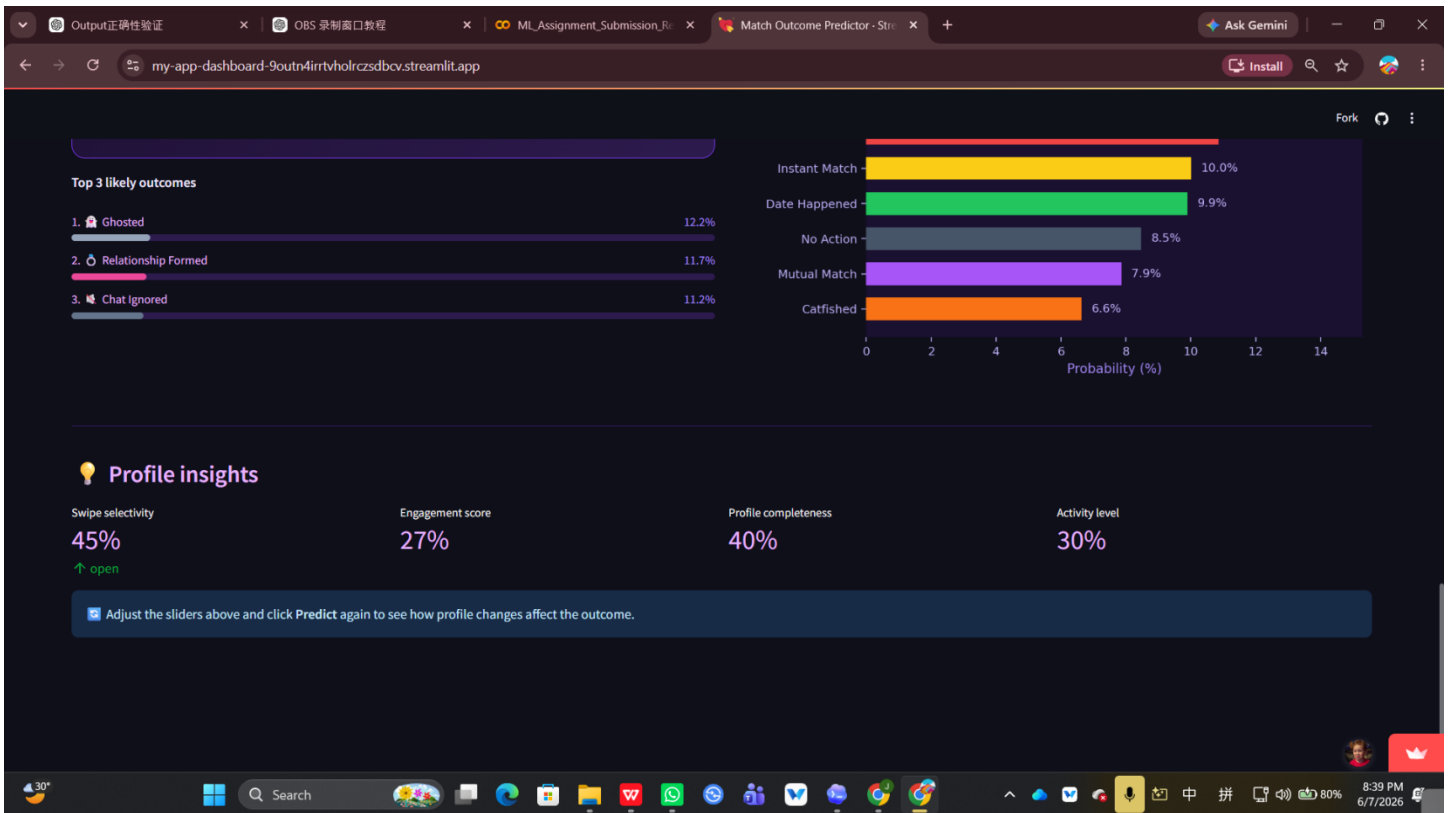
150

0500

Engagement

6





6. Responsible and Ethical Use

Because relationship outcomes involve sensitive social behaviour, the model was framed as an educational prototype rather than a decision-making system. The dashboard warning, low-confidence interpretation, and near-chance performance discussion prevent users from treating the output as personal dating advice.

The dataset is synthetic, so the project avoids making claims about real individuals or groups. The final interpretation focuses on model reliability, data limitations, and professional communication of uncertainty, which supports ethical reporting of machine learning results.

7. Conclusion

Group 5, Team TrueLove successfully developed a complete machine learning workflow for predicting dating-app match outcomes. The project included preprocessing, feature transformation, six-model comparison, GridSearchCV tuning, dimensionality reduction, evaluation, auto-sklearn/FLAML AutoML comparison, and Streamlit prototype deployment.

The project retained the original 10-class target, making the task more rigorous and realistic. The Dummy Baseline achieved 10.22% accuracy, while the best tuned model, Random Forest, achieved 10.15% accuracy. Since this result remains near chance level, the models did not learn meaningful predictive patterns.

The final conclusion is clear: the synthetic dataset lacks sufficient signal for predicting relationship outcomes. This supports the "Garbage In, Garbage Out" principle in machine learning. The Streamlit prototype is educational only, not dating advice.